Tail-Energy Exterior Concentration

Reducing the Trace-to-Spectral Four-Volume Penalty

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Abstract

For a projected cross K with singular values $s_1 \geq \cdots \geq s_r$, define

$$\kappa_4(K) = \frac{\mathrm{e}_4(s_1^2, \ldots, s_r^2)}{s_1^2 s_2^2 s_3^2 s_4^2}.$$

RH-109 used the universal bound $1 \leq \kappa_4 \leq \binom{r}{4}$ to convert a trace exterior moment into a spectral four-volume. This paper replaces the worst combinatorial factor by a perturbation-stable tail-energy bound. If $\widehat{s}_j$ are recent singular values, $\left\|K - \widehat{K}\right\|_2 \leq \delta$, $\ell_j = (\widehat{s}_j - \delta)_+$, $u_j = \widehat{s}_j + \delta$, and

$$S_+ = \left(\left\|\widehat{K}\right\|_F + \sqrt{r}\,\delta\right)^2 - \sum_{j=1}^{4} \ell_j^2,$$

then $\kappa_4(K)$ is bounded by the minimum of $\binom{r}{4}$ and an explicit degree-four polynomial in S_+. Dividing the trace exterior lower bound by this concentration upper bound gives a refined spectral-volume certificate. On the archived fine chain the true concentration never exceeds 2.29093 although $\binom{7}{4} = 35$. The refined trace certificate covers 78, 69, and 55 of 78 updates at thresholds $10^{-8}, 10^{-6}, 10^{-4}$, improving the generic counts 78, 65, and 42. At 10^{-4} it matches the spectral exterior count. No all-level concentration, Stage A, Hilbert–Polya, or Riemann Hypothesis result is claimed.

1 Introduction

RH-109 separated spectral and trace four-volumes, and RH-110 isolated the relative capacity factor [3, 2]. The remaining trace loss is the multiplicity of four-dimensional singular products. A packet of rank seven has 35 exterior coordinates, but the physical spectrum need not distribute mass evenly among them. The scalar κ_4 measures that distribution exactly.

This paper proves a concentration estimate from a quantity already present in the finite-memory calculation: the Frobenius energy outside the first four recent singular directions. The result is useful when lower singular modes carry little energy, and it is sharp at the two universal endpoints.

2 Exterior concentration

Let $a_j = s_j^2$ and $P_4 = a_1 a_2 a_3 a_4$. Since the eigenvalues of $\wedge^4(K^*K)$ are the products $a_{i_1} \cdots a_{i_4}$,

$$\kappa_4 = \frac{\operatorname{tr}(\wedge^4 K^*K)}{\lambda_{\max}(\wedge^4 K^*K)} = \frac{\mathrm{e}_4(a_1, \ldots, a_r)}{P_4}. \tag{1}$$

Thus κ_4 is the stable rank of the exterior Gramian in trace/operator form [1].

Proposition 2.1 (Sharp universal range)**.** *For $s_4 > 0$,*

$$1 \le \kappa_4 \le D_r := \binom{r}{4}.$$

Both bounds are sharp: rank exactly four gives one, and a flat nonzero spectrum gives D_r.

Proof. The numerator of (1) is a sum of D_r nonnegative products, each no larger than P_4, and one term equals P_4. $\square$

3 Tail-energy concentration theorem

Assume $\left\|K - \widehat{K}\right\|_2 \le \delta$ and set $\ell_j = (\widehat{s}_j - \delta)_+$, $u_j = \widehat{s}_j + \delta$. The Frobenius perturbation obeys

$$\|K\|_F \le \left\|\widehat{K}\right\|_F + \sqrt{r}\,\delta.$$

Hence the squared singular energy below the first four modes satisfies

$$S := \sum_{j\ge 5} s_j^2 \le S_+ := \left(\left\|\widehat{K}\right\|_F + \sqrt{r}\,\delta\right)^2 - \sum_{j=1}^{4} \ell_j^2. \tag{2}$$

The right side is replaced by zero if negative.

Theorem 3.1 (Tail-energy concentration upper bound)**.** *If $\prod_{j=1}^4 \ell_j > 0$, then*

$$\kappa_4(K) \le \min\left\{ D_r, \frac{1}{\prod_{j=1}^4 \ell_j^2} \sum_{k=0}^{4} \mathrm{e}_{4-k}(u_1^2, u_2^2, u_3^2, u_4^2) \frac{S_+^k}{k!} \right\}. \tag{3}$$

Proof. Split the elementary symmetric polynomial into terms containing exactly k tail eigenvalues:

$$\mathrm{e}_4(a_1, \dots, a_r) = \sum_{k=0}^{4} \mathrm{e}_{4-k}(a_1, \dots, a_4)\, \mathrm{e}_k(a_5, \dots, a_r).$$

Weyl gives $a_j \le u_j^2$ for the top modes and $P_4 \ge \prod \ell_j^2$. For nonnegative tail eigenvalues, $\mathrm{e}_k(a_5, \dots, a_r) \le S^k/k! \le S_+^k/k!$. Insert these inequalities and combine with Proposition 2.1. $\square$

Corollary 3.2 (Refined trace exterior certificate)**.** *Let κ_+ denote the right side of (3). Then*

$$\frac{s_4(K)}{s_1(K)} \ge \frac{\sqrt{\mathrm{e}_4(\ell_1^2, \dots, \ell_r^2)/\kappa_+}}{(\widehat{s}_1 + \delta)^4}. \tag{4}$$

Proof. The trace exterior norm squared is $\mathrm{e}_4(s_1^2, \dots, s_r^2)$, while the spectral exterior norm squared is the trace divided by κ_4. Use the coordinatewise Weyl lower bounds, Theorem 3.1, and the RH-109 inequality $s_4/s_1 \ge \nu_4$. $\square$

4 Sharpness and interpretation

The flat spectrum shows that the dimension cap in (3) cannot be removed generically. The rank-four spectrum shows that tail energy zero makes the trace and spectral routes identical. Between these endpoints, S_+ records how many lower modes can participate in exterior products and how much weight they can carry.

The theorem is insensitive to singular-vector orientation; this is both its strength and its remaining limitation. A directional tail estimate can be strictly sharper than the Frobenius energy S_+. That question is deferred to the next layer.

Table 1: Fine support counts out of 78.

threshold	generic trace	tail-energy trace	spectral exterior
10^{-8}	78	78	78
10^{-6}	65	69	72
10^{-4}	42	55	55

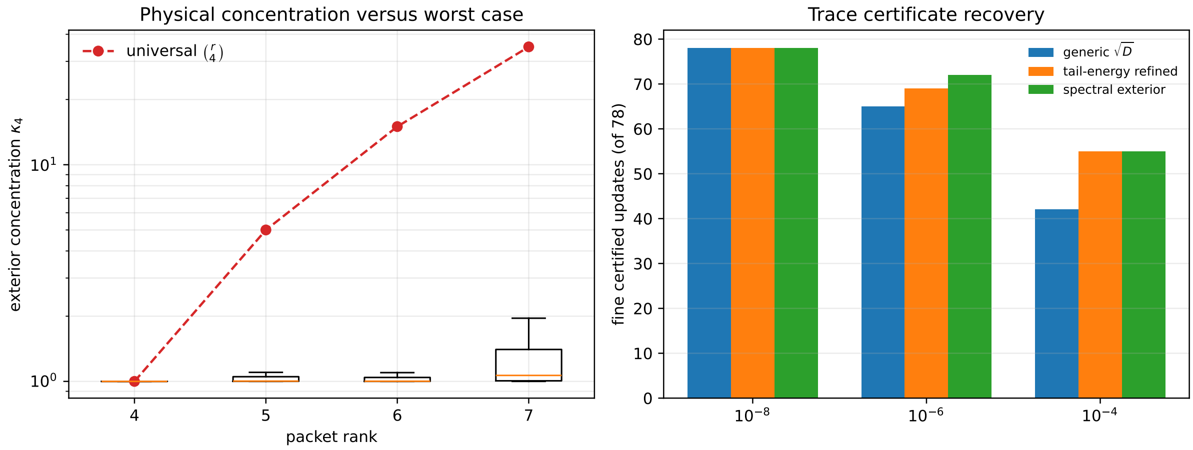

Figure 1: Left: observed concentration is far below the universal exterior dimension. Right: the tail-energy refinement closes most of the generic trace-to-spectral gap.

5 Archived audit

The audit reads the complete recent and full spectra archived by RH-110 for five scales, two channels, and three thresholds. There are 360 records. No true concentration exceeds the bound in (3).

Across the fine records,

$$1.0000000003 \leq \kappa_4 \leq 2.29093,$$

and the refined lower bound is between 1.836 and 5.908 times the generic $D_r^{-1/2}$ trace bound. The minimum refined fine lower bound is 2.42222×10^{-8}, compared with 1.31917×10^{-8} generically. At 10^{-4} the refinement recovers every spectral-volume success; at 10^{-6} three spectral successes remain outside the trace route.

6 Claim boundary and next step

The paper proves a tail-energy concentration theorem and a refined trace certificate. It validates the refinement on the archived finite chain. It does not prove that S_+ or κ_4 obeys an all-level scale law, and it does not provide the missing physical four-volume lower bound.

The next test is whether one should propagate the exterior operator directly rather than through individual singular values. A global wedge-Lipschitz bound is mathematically natural, but its multilinear mixed terms may erase weak four-volume. RH-112 will decide that route quantitatively. No Stage A, Hilbert–Polya operator, zero identification, or Riemann Hypothesis claim is made.

7 Reproducibility

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
```

```
  experiments/build_concentration_audit.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_concentration_audit.py --smoke
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/make_figures.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q -p no:cacheprovider
```

References

[1] Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.

[2] Liang Wang. Finite-memory three-mode capacity. Preprint and software repository, RH-110, 2026.

[3] Liang Wang. Exterior-power fourth-cross support. Preprint and software repository, RH-109, 2026.